

Junbo Koh

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EDUCATION

Seoul National University (SNU)

- M.A. in Educational Technology & Integrated Major in Applied Data Science
- Advisor: [Cheolil Lim](#) & [Yohan Jo](#)

Seoul, South Korea
Mar 2024 - Feb 2026

Gongju National University of Education (GJUE)

- B.S. in Elementary Education, Specializing in Mathematics
- Advisor: Youngmi Cho

Chungnam, South Korea
Mar 2012 - Feb 2016

RESEARCH INTEREST

My research interests focus on “**Human-Centered AI in Education (AIED)**” with a theory-driven approach to addressing real-world educational challenges. I am particularly interested in how AI systems can foster **learner agency** so that students can make their own learning paths and take ownership of their lives, engaging AI as a collaborative partner rather than depending on it.

PUBLICATION

Journal Papers and Conferences

- [j.11] **Learner Experiences of Bounded Artificial Agency in Inquiry Learning: Between Pedagogical Steering and Frustration**
[Junbo Koh](#), Jahyun Jeong, Minsun Cho, Yoomin Kim, Chanhui Kim, and Sunyoung Keum
British Journal of Educational Technology, (Under the Review) [SSCI, Q1, IF 11.1]
- [j.10] **Development and implementation of a generative artificial intelligence-enhanced simulation to enhance problem-solving skills for pre-service teachers**
Jieun Lim, Unggi Lee, [Junbo Koh](#), Yeil Jeong, Yunseo Lee, Gyuri Byun, Haewon Jung, Yoonsun Jang, Sanghyeok Lee, and Jewoong Moon
[Computers & Education](#), 232, 105306. [SCIE/SSCI, Q1, IF 10.5]
- [j.9] **I See You: Teacher Analytics with GPT-4 Vision-powered Observational Assessment**
Unggi Lee, Yeil Jeong, [Junbo Koh](#), Gyuri Byun, Yunseo Lee, Hyunwoong Lee, Seunmin Eun, Jewoong Moon, Cheolil Lim, and Hyeoncheol Kim
[Smart Learning Environments](#), 11(1), 48. [ESCI, Q1, IF 12.1]
- [j.8] **Generative Artificial Intelligence in Educational Game Design: Nuanced Challenges, Design Implications, and Future Research**
Jewoong Moon, Unggi Lee, [Junbo Koh](#), Yeil Jeong, Yunseo Lee, Gyuri Byun, and Jieun Lim
[Technology, Knowledge and Learning](#), 30, 447-459. [ESCI, Q1, IF 3.5]
- [j.7] **Can ChatGPT be a debate partner? Developing ChatGPT-based application “DEBO” for debate education, findings and limitations**
Unggi Lee, Yeil Jeong, [Junbo Koh](#), Gyuri Byun, Yunseo Lee, Youngsun Hwang, Hyeoncheol Kim, and Cheolil Lim
[Educational Technology & Society](#), 27(2), 321-346. [SSCI, Q1, IF 6.0]
- [j.6] **Development of a Teacher Competency Model for AI Digital Textbook-Integrated Instruction**
Kungwook Jeong, Daeun Jeong, Hyejin Lee, and [Junbo Koh \(Corresponding author\)](#)
[Journal of Educational Technology](#), 41(4), 1643-1682. (Top journal of educational technology research in Korea)
- [j.5] **Development of a prototype simulation-based AI system to support physics inquiry learning**
[Junbo Koh](#), Jiwoong Park, Chanhui Kim, Yoomin Kim, and Sunyoung Keum

[The Journal of Educational Information and Media, 31\(2\), 493-517](#). (Top journal of educational technology research in Korea)

- [j.4] **A scoping review of AI utilization and AI convergence education in Korean elementary education**
Kungwook Jeong, **Junbo Koh**, Yeono Son, Minsung Kim, and Sunyoung Keum
[The Journal of Educational Information and Media, 31\(1\), 169-193](#). (Top journal of educational technology research in Korea)
- [j.3] **Learner Perceptions and the Feasibility of Implementing AI Tutors in Distance Universities: Focusing on Improvements to the Current Tutoring System**
Jahyun Jeong, **Junbo Koh**, and Haengsu Shin
[Journal of Lifelong Learning Society, 21\(3\), 82-112](#).
- [j.2] **Designing a mathematics dashboard for data-driven personalized instruction: Focusing on the cognitive diagnostic assessment platform ‘MathCoDi’**
Eunseon Lim, Hyowon Wang, Juhyeon Park, **Junbo Koh**, Cheolil Lim, and Yunjoo Yoo
[Journal of Educational Technology, 41\(2\), 509-546](#). (Top journal of educational technology research in Korea)
- [j.1] **Development of a programming-based instructional program to enhance digital competency of pre-service social studies teachers**
Sang-Il Lee, Bokyoung Go, **Junbo Koh**, Sechang Kim, Woohyung Kim, and Cheolil Lim
[Journal of Educational Technology, 41\(1\), 79-109](#). (Top journal of educational technology research in Korea)
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- [c.18] **Peer and Tutor in One: A Productive Failure-Based Architecture for Multi-Role Conversational Agents in Physics Learning**
Junbo Koh, Minsun Cho, and Sunyoung Keum
AIED 2026: The 27th International Conference on Artificial Intelligence in Education. ([Link](#))
- [c.17] **Rewarding How Models Think Pedagogically: Integrating Pedagogical Reasoning and Thinking Rewards for LLMs in Education**
Unggi Lee, Jiyeong Bae, Jaehyeon Park, Haeun Park, Taejun Park, Younghoon Jeon, Sungmin Cho, **Junbo Koh**, Yeil Jeong, and Gyeonggeon Lee
arXiv preprint arXiv:2601.14560 (Under the Review) ([Link](#))
- [c.16] **OpenLearnLM Benchmark: A Unified Framework for Evaluating Knowledge, Skill, and Attitude in Educational Large Language Models**
Unggi Lee, Sookbun Lee, Heungsoo Choi, Jinseo Lee, Haeun Park, Younghoon Jeon, Sungmin Cho, Minju Kang, **Junbo Koh**, Jiyeong Bae, Minwoo Nam, Juyeon Eun, Yeonji Jung, and Yeil Jeong
arXiv preprint arXiv:2601.13882 (Under the Review) ([Link](#))
- [c.15] **Agents for Agency: Development of AI Agents to Support Agency in Generative AI-Integrated Data Analysis**
Junbo Koh, Woohyeok Kang, Yohan Jo, and Cheolil Lim
ICER 2026: The 26th International Conference on Education Research. ([Link](#))
- [c.14] **Pedagogy-R1: Pedagogical Large Reasoning Model and Well-balanced Educational Benchmark**
Unggi Lee, Jaeyong Lee, Jiyeong Bae, Yeil Jeong, **Junbo Koh**, Gyeonggeon Lee, Gunho Lee, Taekyung Ahn, and Hyeoncheol Kim
CIKM 2025: The 34th ACM International Conference on Information and Knowledge Management. ([Link](#))
- [c.13] **A development of a simulation-based AI system for supporting physics inquiry-based learning**
Junbo Koh, Chanhui Kim, Yoomin Kim, Jahyun Jeong, Minsun Cho, Jiwoong Park, and Sunyoung Keum
AECT 2025: Association for Educational Communications and Technology International Convention. ([Link](#))
- [c.12] **Clarifying AI Education: A Scoping review of AI-Utilization and AI-Convergence Education in Korean Elementary Education**
Kyungwook Jeong, **Junbo Koh**, Yeono Son, Minsung Kim, and Sunyoung Keum
AECT 2025: Association for Educational Communications and Technology International Convention. ([Link](#))

- [c.11] **LLaVA-Docent-V2: Improving Data Quality to Train Large Multimodal Models for Art Appreciation Education**
Unggi Lee, Yoorim Son, Jaeyoon Shin, Gyuri Byun, Yunseo Lee, **Junbo Koh**, Minji Jeon, and Hyeoncheol Kim
ITS 2025: 21th International Conference on Intelligent Tutoring Systems. ([Link](#))
- [c.10] **AIED for Student Agency in the Generative AI Era : A Scoping Review**
Junbo Koh
ICoME 2025: The 23rd International Conference for Media in Education. ([Link](#))
- [c.9] **Development of an Agentic AI for Novice Instructional Designers**
Unggi Lee, Hyeeseong Kwon, **Junbo Koh**, Yeil Jeong, Gyuri Byun, Junseo Kim, Minji Jeon, and Cheolil Lim
ICER 2025: The 25th International Conference on Education Research. ([Link](#))
- [c.8] **Evaluating the Usability of an LMS-based Educational Metaverse: A Case Study of Wonderverse**
Juhyeon Park, Minhee Lim, **Junbo Koh**, Jahyun Jeong, Junseo Kim, and Cheolil Lim
ICoME 2024: The 22nd International Conference for Media in Education. ([Link](#))
- [c.7] **Design and development of an AI system for generating instructional materials based on instructional design principles: A case study of Korean medicine education**
Cheolil Lim, **Junbo Koh**, Hoetaek Rah, Gyuri Byun, and Hyesung Kwon
2026 Spring Conference of the Korean Society for Educational Technology.
- [c.6] **Development of a teacher competency model for AI digital textbook-integrated instruction: Based on the High-Touch High-Tech approach**
Kungwook Jeong, Daeun Jeong, Hyejin Lee, and **Junbo Koh (Corresponding author)**
2025 Joint Fall Conference of the Korean Association for Educational Information and Media and the Korean Society for Educational Technology
- [c.5] **Development of Pedagogical Reasoning and a Pedagogical Benchmark for Large Language Models**
Unggi Lee, Jaeyong Lee, **Junbo Koh**, and Gyeonggeon Lee
2025 Spring Conference of the Korean Society for Educational Technology.
- [c.4] **Exploring pre-service teacher education approaches for enhancing AI digital textbook utilization competencies based on educational needs analysis**
Cheolil Lim, **Junbo Koh**, Juhyun Park, Seongkyung Park, Junseo Kim, and Kungwook Jeong
2025 Spring Conference of the Korean Society for Educational Technology.
- [c.3] **Development of a simulation-based AI system to support physics inquiry processes**
Junbo Koh, Jiwoong Park, Chanhui Kim, Yoomin Kim, and Sunyoung Keum
2025 Spring Conference of the Korean Association for Educational Information and Media
- [c.2] **Development of a large language model-based ISD agent to support novice instructional designers**
Hyesung Kwon, **Junbo Koh**, Yeil Jeong, Gyuri Byun, Junseo Kim, Cheolil Lim, Unggi Lee, and Minji Jeon
2025 Spring Conference of the Korean Association for Educational Information and Media
- [c.1] **Analyzing research trends on educational practices of AI use and AI integration in Korean elementary education: A scoping review**
Kungwook Jeong, **Junbo Koh**, Yeono Son, Minsung Kim, and Sunyoung Keum
2024 Joint Fall Conference of the Korean Association for Educational Information and Media and the Korean Society for Educational Technology

Posters, Demos, and Workshop Papers

- [p.5] **Agent for Agency: Developing an AI Agent to Support Student Agency in Generative AI-Supported Learning Environments**
Junbo Koh, Jiyeon Han, Woohyeok Kang, Cheolil Lim, and Yohan Jo
LAK26: The 16th International Learning Analytics and Knowledge Conference

- [p.4] **Does Collaborative Learning Theory Apply to AI?: Exploring collaboration pattern for LLM-agents through Group Communication Analysis**
Junbo Koh, Taekyun Yoo, Juyeon Lee, and Yunho Maeng
ICER 2024: The 24th International Conference on Education Research.
- [p.3] **A Study on the Development of Online Education Content Manual for Korean Medicine Using RPISD Model**
 Cheolil Lim, Sumin Hong, Eunseon Lim, Taeyeon Eom, Kyungwook Jeong, **Junbo Koh**, Yoomin Kim, and Jiseong Hong
ICER 2024: The 24th International Conference on Education Research.
- [p.2] **A study on teacher education university faculty members' perceptions of the functions and use of AI digital textbooks**
 Cheolil Lim, Seongkyung Park, Juhyun Park, **Junbo Koh**, Junseo Kim, and Kungwook Jeong
2025 Spring Conference of the Korean Society for Educational Technology.
- [p.1] **Development of an educational program for enhancing digital education competencies of pre-service geography teachers**
 Bokyoung Go, **Junbo Koh**, Sechang Kim, Woohyung Kim, Sang-II Lee, and Cheolil Lim
2024 Joint Fall Conference of the Korean Association for Educational Information and Media and the Korean Society for Educational Technology
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- [d.1] **Slidecaster: Human-AI Collaborative Authoring of Pedagogically Grounded Micro-Learning Videos**
Junbo Koh, Hoetaek Rah, Hyeseong Kwon, Gyuri Byun, and Cheolil Lim
AIED 2026: The 27th International Conference on Artificial Intelligence in Education. ([Link](#))
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- [w.1] **Generative agent for teacher training: Designing educational problem-solving simulations with large language model-based agents for pre-service teachers**
 Unggi Lee, Sanghyeok Lee, **Junbo Koh**, Yeil Jeong, Haewon Jung, Gyuri Byun, Yunseo Lee, Haewon Jung, Jewoong Moon, Jieun Lim, and Hyeoncheol Kim
NeurIPS 2023 Workshop on Generative AI for Education (GAIED)

EXPERIENCES

Researcher (Learning Sciences Research Institute , SNU)	Seoul, South Korea Mar 2024 - Present
AI Digital Education Lead Teacher (Seoul Metropolitan Office of Education) - Designed and delivered AI/edtech-integrated lessons, moving beyond tool usage toward learning experience design - Guided school-wide AI transformation strategy and fostered peer collaboration among fellow educators	Seoul, South Korea May 2026 - Present
AI Convergence Education Lead Teacher (Seoul Metropolitan Office of Education) - Researched and curated educational content related to AI convergence education - Documented and shared teaching cases from AI integrated classroom experiences	Seoul, South Korea May 2021 - Feb 2022
Arts, Culture & Physical Education Department Head (Seoul Haenghyun Elementary School) - Coordinated arts, culture and PE education programs across the department - Recruited and supervised instructors for various educational programs - Organized fitness assessment and field trip programs throughout the academic year	Seoul, South Korea Mar 2022 - Feb 2023
6th Grade Head Teacher (Seoul Haenghyun Elementary School) - Managed 6th grade administrative operations and coordinated grade-level activities - Organized curriculum planning for 6th grade classes - Coordinated teacher training programs for grade-level faculty	Seoul, South Korea Mar 2021 - Feb 2022

ACADEMIC SERVICES

Journal Reviewer

- Computers & Education (Elsevier, SSCI), 2 manuscripts / 3 review rounds, 2025-2026
- Computers and Education Open (Elsevier), 1 manuscript, 2026

TEACHING

Elementary School Teacher (Seoul Metropolitan Office of Education)

Seoul, South Korea
Mar 2016 - Present

- Designed instructional materials and lesson plans for students
- Guided student behavior and social development
- Contributed to curriculum development for grade-level programs

AWARDS & GRANTS

Junior Scholar Fellowship

Feb 2026

- \$1300 merit-based scholarship awarded by [BK21 FOUR, SNU](#). Awarded to an emerging scholar for active research contributions and demonstrating high academic potential during 2025.

Jr. Priming Award

June 2025

- \$1300 merit-based scholarship awarded by [BK21 FOUR, SNU](#). Awarded to emerging scholars based on research proposal evaluation, providing academic support.

Graduate Student Learning Sciences Research Scholarship

May 2025

- \$1300 merit-based scholarship awarded by [Learning Sciences Research Institute, SNU](#). Awarded to interdisciplinary educational projects focusing on learning sciences.

CERTIFICATIONS

Class 1 Regular Teacher Certificate

- Seoul Metropolitan Office of Education - No. 71-46264

Class 2 Regular Teacher Certificate

- Gongju National University of Education - No. 17427

SKILLS

Programming: Python, C++

AI & Data Science: PyTorch, LangGraph, LangChain, scikit-learn, Pandas, Matplotlib

Frameworks: Streamlit, Django, React

Other: Git, AWS (EC2, S3, Lambda, DynamoDB)

Research: Design Research, Interview design, Qualitative coding, Statistical analysis